

Microeconomic Analysis

Topics on Information Economics

HT 2025

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Topics on Information Economics

I Hidden Action in Principal-Agent Problems

II Hidden Information in Principal-Agents Problems

Principal-Agent Problem - Hidden Information

Setting:

A risk neutral principal hires a risk averse agent for a one-time project.

The agent exerts **observable** effort $e \geq 0$.

Gross profit are deterministic: $\pi(e)$, with $\pi(0) = 0$, $\pi' > 0$, $\pi'' < 0$.

Information:

After the contract is signed, the agent discovers her disutility of effort:

Agent's utility: $u(w, e, \theta) = v(w - g(e, \theta))$, with $v' > 0$, $v'' \leq 0$.

Here, θ is **the state of nature** that is realised *after* the contract is signed, and is *privately* observed by the agent;

ex ante, when the contract is signed, there is symmetric information.

Principal-Agent Problem - Hidden Information

$$u(w, e, \theta) = v(w - g(e, \theta))$$

The function g measures the cost of effort in monetary terms.

We assume:

$$g(0, \theta) = 0, \quad \forall \theta \quad \text{No disutility if no effort}$$

$$g_e(e, \theta) > 0, \quad g_{ee}(e, \theta) < 0, \quad \forall \theta \text{ and } e > 0 \quad \text{Increasing and convex cost function}$$

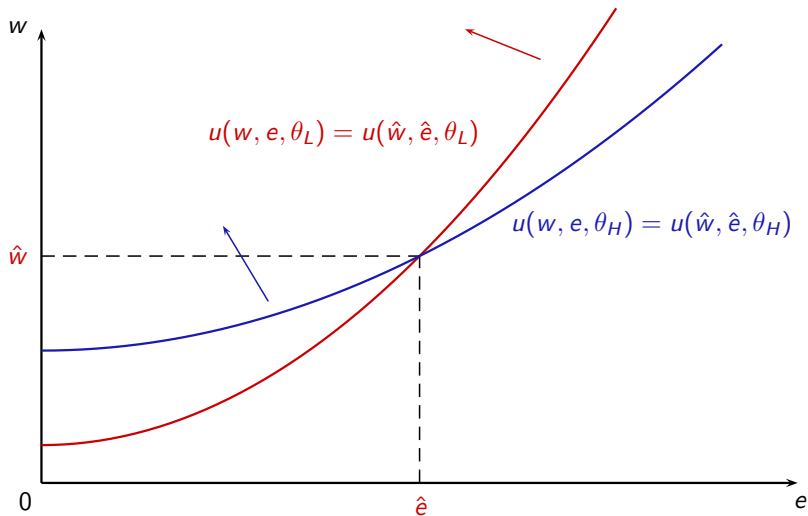
$$g_\theta(e, \theta) < 0, \quad g_{e\theta}(e, \theta) < 0, \quad \forall \theta \text{ and } e > 0 \quad \text{Single-crossing property}$$

Agent has *ex ante* reservation utility $\bar{u}$.

For simplicity we will focus on the case of two states:

$$\theta \in \{\theta_L, \theta_H\}; \quad \theta_L < \theta_H; \quad \lambda = \Pr[\theta = \theta_H].$$

Single-Crossing Property



Principal-Agent Problem - Hidden Information

Contracts:

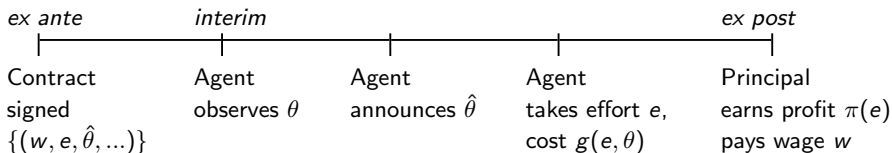
The contract specifies the wage w and the effort e .

Contracts could be complex, in particular, the details of the contract can depend on an announcement, $\hat{\theta}$, that the agent makes after observing θ .

Objectives of the contract:

- Risk neutral owner insures the manager against fluctuations in income.
- Managerial effort should respond to true disutility of effort θ .

Timing:



Benchmark I: the state θ is observable

If θ is observable, a contract can specify (w, e) as a function of θ :

$$\{(w_H, e_H), (w_L, e_L)\}$$

Principal's problem:

$$\begin{aligned} \max_{(w_L, e_L), (w_H, e_H)} \quad & \lambda [\pi(e_H) - w_H] + (1 - \lambda) [\pi(e_L) - w_L] \\ \text{s.t.} \quad & \lambda v(w_H - g(e_H, \theta_H)) + (1 - \lambda) v(w_L - g(e_L, \theta_L)) \geq \bar{u} \end{aligned}$$

Let $\gamma \geq 0$ be the Kuhn-Tucker multiplier on the the agent's *ex ante* Participation constraint.

The FOCs for the solution to this **first best** problem are :

$$\lambda = \gamma \lambda v'(w_H - g(e_H, \theta_H)) \quad [FOC_{w_H}]$$

$$(1 - \lambda) = \gamma (1 - \lambda) v'(w_L - g(e_L, \theta_L)) \quad [FOC_{w_L}]$$

$$\lambda \pi'(e_H) = \gamma \lambda v'(w_H - g(e_H, \theta_H)) g_e(e_H, \theta_H) \quad [FOC_{e_H}]$$

$$(1 - \lambda) \pi'(e_L) = \gamma (1 - \lambda) v'(w_L - g(e_L, \theta_L)) g_e(e_L, \theta_L) \quad [FOC_{e_L}]$$

Benchmark I: the state θ is observable

The first two FOCs imply that

$$\gamma v'(w_H^* - g(e_H^*, \theta_H)) = \gamma v'(w_L^* - g(e_L^*, \theta_L)) = 1$$

In particular:

- $\gamma > 0$, so the Participation constraint binds (by complementary slackness).
- so v' is equalised across states and we have **optimal risk-sharing**:

$$w_H^* - g(e_H^*, \theta_H) = w_L^* - g(e_L^*, \theta_L) \quad \text{since } v' > 0.$$

- so v is equalised across states, and

$$v(w_H^* - g(e_H^*, \theta_H)) = v(w_L^* - g(e_L^*, \theta_L)) = \bar{u}$$

Benchmark I: the state θ is observable

Now consider effort levels in the two states:

Substituting $\gamma v' = 1$ into the last two FOCs gives

$$\begin{aligned}\pi'(e_H^*) &= g_e(e_H^*, \theta_H) \\ \pi'(e_L^*) &= g_e(e_L^*, \theta_L)\end{aligned}$$

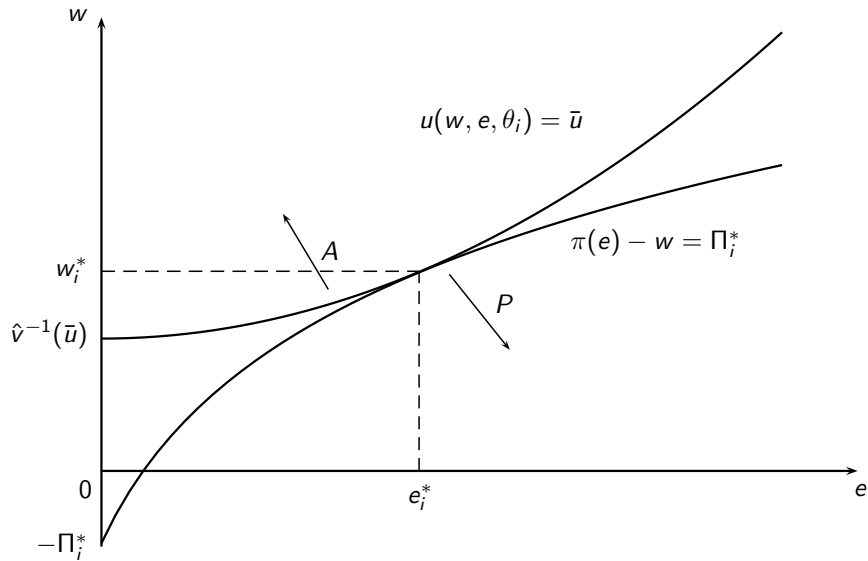
which shows that optimal effort levels equate marginal benefits with marginal costs; **efficient effort**.

Note that $\pi'(e_i^*) = g_e(e_i^*, \theta_i)$ and $v(w_i^* - g(e_i^*, \theta_i)) = \bar{u}$ uniquely determine the contract (w_i^*, e_i^*) .

The profits achieved by the owner are:

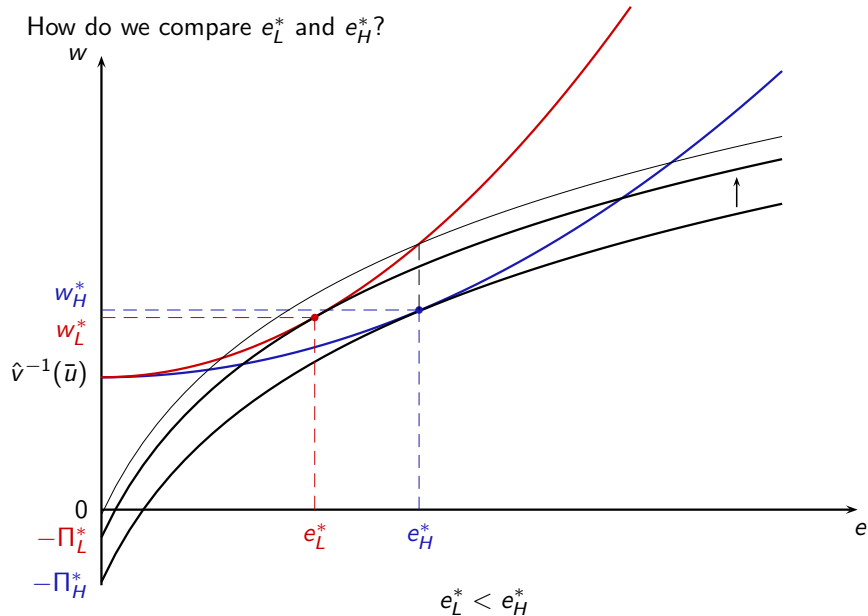
$$\Pi_i^* = \pi(e_i^*) - w_i^* = \pi(e_i^*) - v^{-1}(\bar{u}) + g(e_i^*, \theta_i)$$

Benchmark I: the state θ is observable



Benchmark I: the state θ is observable

How do we compare e_L^* and e_H^* ?



Benchmark II: the agent is risk neutral

As usual, if the agent is risk neutral, the **first best** can be achieved by 'selling the project' to the agent.

Take $v(w) \equiv w$.

If the agent keeps all the profits, she chooses effort to maximise

$$\pi(e) - g(e, \theta_L) \quad \text{or} \quad \pi(e) - g(e, \theta_H)$$

Hence $\pi'(e_i^*) = g_e(e_i^*, \theta_i)$ as before.

Her expected net surplus *ex ante* is

$$(1 - \lambda) [\pi(e_L^*) - g(e_L^*, \theta_L)] + \lambda [\pi(e_H^*) - g(e_H^*, \theta_H)] - \bar{u}$$

and this is what the principal charges the agent for the project.

$$w(e) \equiv \pi(e) - [(1 - \lambda) [\pi(e_L^*) - g(e_L^*, \theta_L)] + \lambda [\pi(e_H^*) - g(e_H^*, \theta_H)] - \bar{u}]$$

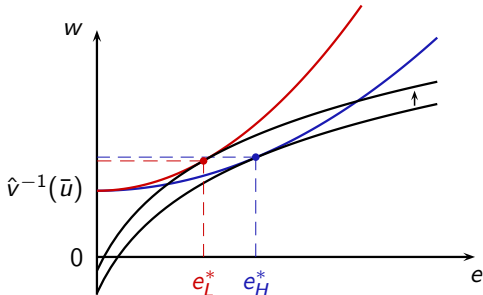
State θ unobservable; manager risk averse

When θ is private information:

Conflict between risk sharing and effort elicitation.

The principal could ask the manager to report the state, but the first best contract is not incentive compatible:

- in state θ_H , the agent would pretend that θ_L had occurred.



The revelation principle

The first step is to consider what are the possible contracts.

Imagine a (possibly very complex) contract that the agent finds acceptable, such that:

- in state θ_H , the agent exerts effort e_H , resulting in a wage $w(e_H)$, and
- in state θ_L , the agent exerts effort e_L resulting in a wage $w(e_L)$.

The agent makes these choices because

$$v(w(e_H) - g(e_H, \theta_H)) \geq v(w(e) - g(e, \theta_H))$$

for any other choice of e in state θ_H , and similarly in state θ_L .

The revelation principle

Now suppose that the principal offers this alternative contract to the agent:

*“When you find out, you tell me if the state is θ_H or θ_L ;
if you say θ_H , then you will put in effort e_H and receive a wage $w(e_H)$, and
if you say θ_L , then you will put in effort e_L and receive a wage $w(e_L)$.”*

If the agent found the original contract acceptable, the agent will find this alternative contract acceptable. Moreover, it will be optimal for the agent to report the state truthfully.

The principal's expected net profit from the alternative contracts is the same as from the original one.

$$\lambda[\pi(e_H) - w(e_H)] + (1 - \lambda)[\pi(e_L) - w(e_L)]$$

So we can restrict the type of contract offered to a **direct mechanism** that induces **truth-telling** without loss of generality.

The revelation principle

We appeal to the **Revelation Principle**, which says that the principal can restrict attention to the set of contracts of the form:

- 1 After the *interim* stage, the agent makes an announcement $\hat{\theta}$ about the state.
- 2 The contract specifies an outcome $(w(\hat{\theta}), e(\hat{\theta}))$ for each possible $\hat{\theta}$.
- 3 In every state θ , truth-telling is optimal for the agent. (Incentive-compatible)

The agent's Incentive Compatibility (truth-telling) constraints are:

$$\begin{aligned}w_H - g(e_H, \theta_H) &\geq w_L - g(e_L, \theta_H) \\w_L - g(e_L, \theta_L) &\geq w_H - g(e_H, \theta_L)\end{aligned}$$

State θ unobservable; manager (infinitely) risk averse

For simplicity, assume that the agent is *infinitely risk averse*,

- In a state-contingent diagram, this correspond to *Leontief* indifference curves:

the agent's *ex ante* expected utility is her lowest *ex post* utility.

The *ex ante* participation constraint becomes:

$$\begin{aligned}w_H - g(e_H, \theta_H) &\geq v^{-1}(\bar{u}) \\w_L - g(e_L, \theta_L) &\geq v^{-1}(\bar{u})\end{aligned}$$

So the problem can be written as:

$$\begin{aligned}\max_{w_H, e_H, w_L, e_L} \quad & \lambda[\pi(e_H) - w_H] + (1 - \lambda)[\pi(e_L) - w_L] \quad s.t. \\(PC_H) \quad & w_H - g(e_H, \theta_H) \geq v^{-1}(\bar{u}) \\(PC_L) \quad & w_L - g(e_L, \theta_L) \geq v^{-1}(\bar{u}) \\(IC_H) \quad & w_H - g(e_H, \theta_H) \geq w_L - g(e_L, \theta_H) \\(IC_L) \quad & w_L - g(e_L, \theta_L) \geq w_H - g(e_H, \theta_L)\end{aligned}$$

State θ unobservable; manager (infinitely) risk averse

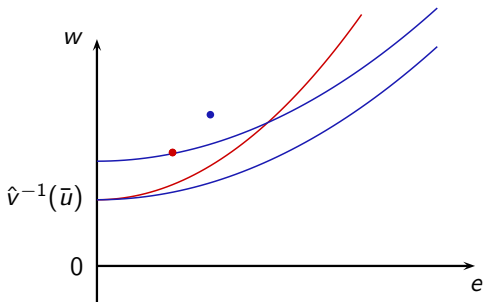
Lemma 1: PC_H

We can ignore the PC_H constraint: $w_H - g(e_H, \theta_H) \geq v^{-1}(\bar{u})$

Proof:

If the constraints (PC_L) and (IC_H) are satisfied, it must be that:

$$w_H - g(e_H, \theta_H) \geq w_L - g(e_L, \theta_H) \geq w_L - g(e_L, \theta_L) \geq v^{-1}(\bar{u})$$



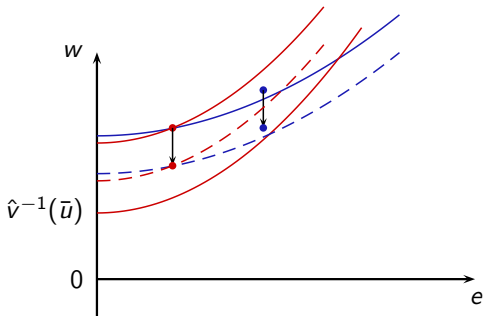
State θ unobservable; manager (infinitely) risk averse

Lemma 2: PC_L

PC_L binds: $w_L - g(e_L, \theta_L) = v^{-1}(\bar{u})$

Proof:

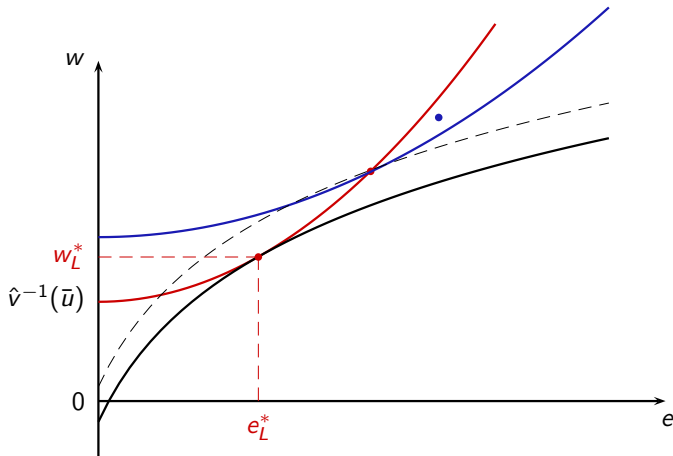
If $w_L - g(e_L, \theta_L) > v^{-1}(\bar{u})$, consider $\hat{w}_L = w_L - \epsilon$, $\hat{w}_H = w_H - \epsilon$. Then, IC_H , IC_L still satisfied, and if ϵ small enough, PC_L satisfied. But the principal is paying less.



State θ unobservable; manager (infinitely) risk averse

Lemma 3

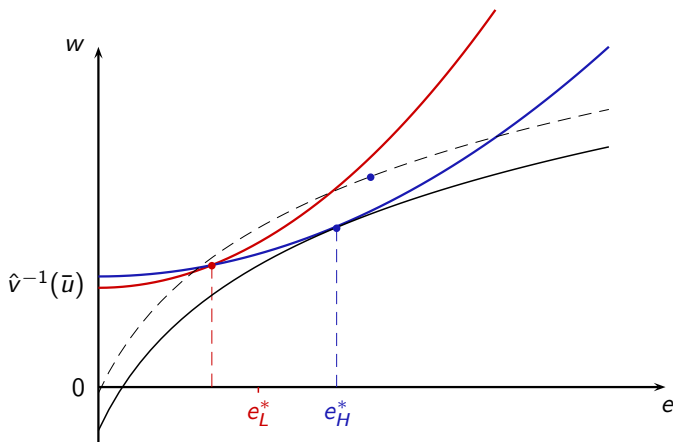
In any optimal contract $\hat{e}_L \leq e_L^*$.



State θ unobservable; manager (infinitely) risk averse

Lemma 4

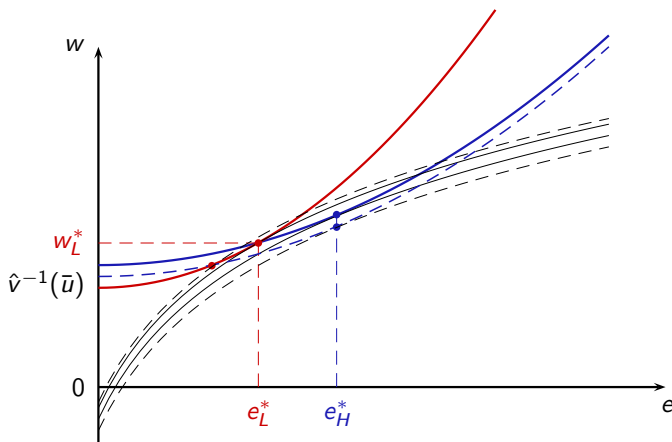
In any optimal contract $\hat{e}_H = e_H^*$



State θ unobservable; manager (infinitely) risk averse

Lemma 5

In any optimal contract $\hat{e}_L < e_L^*$



State θ unobservable; manager (infinitely) risk averse

To sum up:

The optimal contract $(\hat{w}_H, \hat{e}_H)$, $(\hat{w}_L, \hat{e}_L)$, with unobservable θ and infinitely risk averse agent satisfies:

- $\hat{e}_H = e_H^*$: No distortion at the top.
- $\hat{e}_L < e_L^*$: Distortion at the bottom.
- Imperfect risk sharing: $u_H > \bar{u} = u_L$
- Principal loses

State θ unobservable; manager (infinitely) risk averse

Comment:

We assumed that the agent is **infinitely risk averse**,

Hence, the *ex ante* participation constraint becomes:

$$\begin{aligned}w_H - g(e_H, \theta_H) &\geq v^{-1}(\bar{u}) \\w_L - g(e_L, \theta_L) &\geq v^{-1}(\bar{u})\end{aligned}$$

- These are the same as the Participation constraints of an agent who signs a contract only *after* she observes θ , i.e. at the *interim* stage. (**Monopolistic Screening!**)

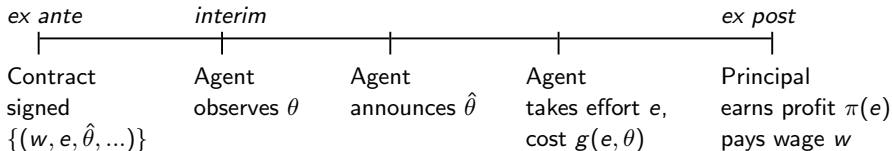
Infinitely Risk Averse
ex ante Participation Constraint



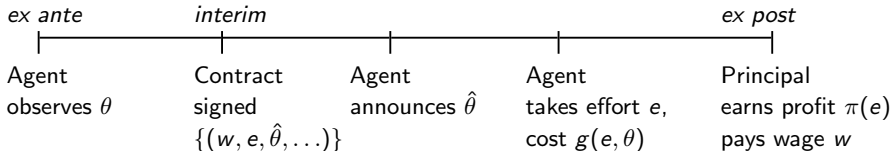
Finitely Risk Averse
interim Participation Constraint

Monopolistic Screening

Principal-agent Hidden Information:



Monopolistic Screening:



Example: Monopolistic Screening

Two types of rail traveller: L and H ; λ = fraction of H -types.

- Type H have utility from travelling: $u_H(m, q) = m + \theta_H q$
- Type L have utility from travelling: $u_L(m, q) = m + \theta_L q$

where m is net money income, q is quality of travel, and θ is a taste parameter with $\theta_H > \theta_L > 0$. (Will also need $\lambda < \theta_L/\theta_H$.)

Suppose that the cost per passenger of providing quality q is $q^2/2k$.

The monopolist charges $p(q)$ when offering quality q .

Example: Monopolistic Screening

First Best:

If θ were public information, the monopoly railroad company would solve:

$$\begin{aligned} \max_{q_i, p_i} \quad & p_i - q_i^2/2k \quad s.t. \\ & \theta_i q_i - p_i \geq 0 \quad (PC) \end{aligned}$$

Hence the monopoly would choose qualities:

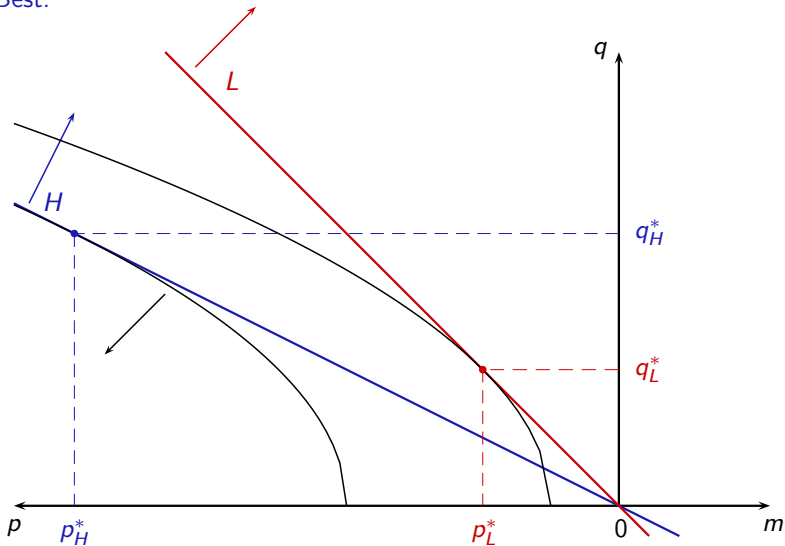
$$q_H^* = k\theta_H, \quad q_L^* = k\theta_L$$

and charge prices:

$$p_H^* = \theta_H q_H^*, \quad p_L^* = \theta_L q_L^*$$

Example: Monopolistic Screening

First Best:



Example: Monopolistic Screening

Suppose that θ is private information.

The monopolist's problem is:

$$\max_{(q_L, p_L), (q_H, p_H)} \lambda [p_H - q_H^2/2k] + (1 - \lambda) [p_L - q_L^2/2k]$$

such that the Participation constraints hold:

$$\theta_H q_H - p_H \geq 0 \quad (PC_H)$$

$$\theta_L q_L - p_L \geq 0 \quad (PC_L)$$

and the Incentive Compatibility (truth-telling) constraints hold:

$$\theta_H q_H - p_H \geq \theta_H q_L - p_L \quad (IC_H)$$

$$\theta_L q_L - p_L \geq \theta_L q_H - p_H \quad (IC_L)$$

Note that (IC_H) and (PC_L) imply (PC_H) hold as a strict inequality whenever $\hat{q}_L > 0$.

Example: Monopolistic Screening

$$\begin{array}{ll} \theta_H q_H - p_H \geq 0 & (PC_H) \quad \theta_H q_H - p_H \geq \theta_H q_L - p_L \quad (IC_H) \\ \theta_L q_L - p_L \geq 0 & (PC_L) \quad \theta_L q_L - p_L \geq \theta_L q_H - p_H \quad (IC_L) \end{array}$$

- Fix the *difference* between p_H and p_L . If (IC_H) and (IC_L) hold, then they will still do so with higher p_H and p_L . So, increase p_L (and p_H) until (PC_L) binds

L-types are indifferent between travelling and not, and given that, H-types definitely want to travel.

- Fix p_L . If (IC_L) holds, then it will still do so with higher p_H . So, increase p_H until (IC_H) binds .

H-types are indifferent between travelling Business and Tourist Class, and given that, L-types definitely want to travel Tourist rather than Business.

Example: Monopolistic Screening

$$\begin{array}{ll} \theta_H q_H - p_H \geq 0 & (PC_H) \quad \theta_H q_H - p_H = \theta_H q_L - p_L \quad (IC_H) \\ \theta_L q_L - p_L = 0 & (PC_L) \quad \theta_L q_L - p_L \geq \theta_L q_H - p_H \quad (IC_L) \end{array}$$

- From equality (PC_L) , we obtain $p_L = \theta_L p_L$.
- And now from equality (IC_H) , we obtain $p_H = \theta_H q_H - (\theta_H - \theta_L)q_L$.

Substitute these into the monopolist's maximisation problem ...

$$\max_{q_H, q_L} \lambda[\theta_H q_H - (\theta_H - \theta_L)q_L - q_H^2/2k] + (1 - \lambda)[\theta_L q_L - q_L^2/2k]$$

$$FOC_{q_H} : \quad \lambda(\theta_H - q_H/k) = 0$$

$$FOC_{q_L} : \quad -\lambda(\theta_H - \theta_L) + (1 - \lambda)(\theta_L - q_L/k) = 0$$

Example: Monopolistic Screening

Solving the relaxed (ignoring IC_L) problem leads to:

$$\hat{q}_H = k\theta_H = q_H^*$$

$$\hat{q}_L = k\theta_L - k(\theta_H - \theta_L)\lambda/(1 - \lambda) < q_L^*$$

$$\hat{p}_H = \theta_H \hat{q}_H - (\theta_H - \theta_L) \hat{q}_L < p_H^*$$

$$\hat{p}_L = \theta_L \hat{q}_L < p_L^*$$

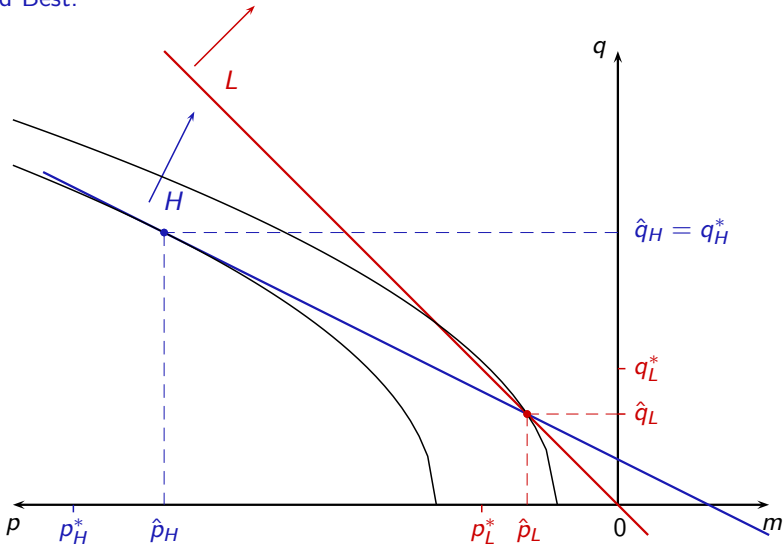
Note that $\lambda < \theta_L/\theta_H$ ensures that $\hat{q}_L > 0$.

Also, note that $\hat{p}_H - \hat{p}_L = \theta_H(\hat{q}_H - \hat{q}_L)$, and so (IC_L) does indeed hold as a strict inequality.

Finally, what happens as $\lambda/(1 - \lambda)$ rises?

Example: Monopolistic Screening

Second Best:



To Sum Up: Monopolistic Screening

At the **second best**,

- $\hat{q}_H = q_H^*$ (no distortion for Business Class),
- $\hat{q}_L < q_L^*$ (poor food & no leg-room in Tourist Class).
- $\hat{p}_H < p_H^*$ (lower, to induce Business travellers not to mimic Tourists), and
- $\hat{p}_L < p_L^*$ (lower, to induce Tourists to travel with lower quality).

In terms of payoffs:

- L traveller indifferent
- H traveller strictly better off
- Monopolist worse off.